Lecture 3

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Sliding mode control and its application

Controllability

- The theory of controllability was proposed in 1960 by R. Kalman.

Definition of Controllability

The linear system with state differential equation

$$\dot{x}(t) = A(t)x(t) + B(t)u(t)$$

is said to be completely controllable if it is possible to construct an unconstrained control signal that will transfer an initial state at any initial time t_0 to any final state $x(t_1) = x_1$ in a finite time interval $t_1 - t_0$.

When we say that the system can be transferred from one state to another, we mean that there exists a piecewise continuous input $u(t), t_0 \leq t \leq t_1$, which brings the system from one state to the other.

A necessary and sufficient condition for controllability in the s plane is that no single pole of the system is cancelled by a zero in all of the elements of the transfer function matrix $[sI - A]^{-1}B$. If such a cancellation occurs, the system cannot be controlled in the direction of the cancelled mode.

Controllability of linear time-invariant Systems

Consider the linear time-invariant system

$$\dot{x}(t) = Ax(t) + Bu(t), \qquad x(t) \in \mathbb{R}^n.$$

This system is controllable if and only if

$$\texttt{rank}[B, AB, \cdots, A^{n-1}B] = n.$$

The controllability matrix is defined as

$$\mathcal{C} = \begin{bmatrix} B & AB & \cdots & A^{n-1}B \end{bmatrix}$$

Observability

Definition of Observability

Let $y(t; t_0, x_0, u)$ denote the response of the output variable $y(t)$ of the linear differential system

$$\dot{x}(t) = A(t)x(t) + B(t)u(t),$$
$$y(t) = C(t)x(t),$$

to the initial state $x(t_0) = x_0$. Then the system is called completely observable if for all t_1 there exists a t_0 with $-\infty < t_0 < t_1$ such that

$$y(t; t_0, x_0, u) = y(t; t_0, x_0^{'}, u), \qquad t_0 \leq t \leq t_1,$$

for all $u(t)$, $t_0 \leq t \leq t_1$, implies $x_0 = x_0^{'}$.

The system is observable over a time period $[t_0 \ t_1]$ if it is possible to determine uniquely the initial state $x(t_0) = x_0$ from the knowledge of the output $y(t)$ over $[t_0 \ t_1]$.
The complete state of the system is known if the initial state x_0 is known.

Observability of Linear Time-Invariant System

Theorem

Consider the linear time-invariant system

$$\begin{aligned} \dot{x}(t) &= Ax(t) + Bu(t), \\ y(t) &= Cx(t), \end{aligned}$$

is completely observable if and only if the row vectors of the Observability matrix

$$Q = \begin{pmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{pmatrix}$$

span the n-dimensional space.

Controllability and Reachability in discrete-time system

Reachability

The system is reachable if the state can be transferred from the zero state at any initial time t_0 to any terminal state $x(t_1) = x_1$ within a finite time $t_1 - t_0$.

Controllability

The system is controllable if the state can be transferred from any state at any initial time t_0 to any terminal state $x(t_1) = 0$ within a finite time $t_1 - t_0$.

In a continuous-time system, controllability and reachability are the same.

Controllability and Reachability

Controllability Need Not Imply Reachability

Consider the system

$$x(k+1) = Ax(k).$$

If A is a nilpotent matrix, then the system states

$$x(k) = A^k x(0)$$

can be driven to zero from any initial state without any input. Such a system is controllable but not reachable.

For one more example of such a system consider

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \qquad b = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

then,

$$\begin{bmatrix} b & Ab \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$$

is singular.

Controllability and Reachability

Controllability Need Not Imply Reachability

So, the arbitrary state, say $[1\ 1]^\top$, can not be reached from the zero state no matter what the input is. On the other hand, any nonzero initial state can be returned to zero. For $x_1(0) = \alpha$ and $x_2(0) = \beta$, let $u(0) = -(\alpha + \beta)$, then

$$x(1) = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} (-\alpha - \beta) = 0.$$

Advantages of Linear Systems :

Almost all systems present in nature are nonlinear. It is difficult to analyze a nonlinear system, so we approximate it as a linear system for our convenience and to utilize the following advantages it offers:

- Can scale and add solutions to get new solutions.
- Response to any input can be completely determined by response to a basic set of inputs, e.g., $\delta(t)$, $sin(\omega t + \phi)$, e^{-st}.
- Solution set forms a vector space. It is enough to find a basis of the solution. (Linear algebra toolkit can be used).
- The existence and uniqueness of the solution is always guaranteed.

Limitations of Linearization:

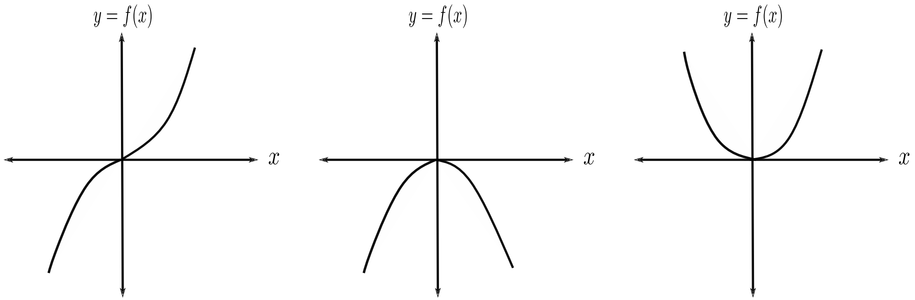


Figure: Non Linear Systems

- Linearize all three examples about the origin.
- Linearized system is given by straight line ($y = 0$).
- Systems are quite different, but linearization gives the same behaviour.

Notations

1. Set of Real no: $\mathbb{R}$
2. Let $n > 0$ is integer, set of real numbers $x_1, x_2, ...x_n$ where $x_i \in \mathbb{R}$
3. Vector or a point in $\mathbb{R}^n$ ($\mathbb{R} \times \mathbb{R} \times \mathbb{R}.... \times \mathbb{R} = \mathbb{R}^n$) space is

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}_{n\times 1}$$

Euclidean Space

1. Add vectors
2. Scalar multiplication
3. Zero vector in $\mathbb{R}^n$ is $\begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}_{n\times 1}$. Also called a trivial vector or the origin of the $\mathbb{R}^n$ space.

Nonlinear System

Equilibrium point

Consider a nonlinear system given by

$$\dot{x} = f(t, x(t), u(t)) \qquad f : \mathbb{R}^+ \times \mathbb{R}^n \times \mathbb{R} \rightarrow \mathbb{R}^n \tag{1}$$

A vector $x_e \in \mathbb{R}^n$ is said to be an equilibrium point of (1) if

$$f(t, x_e) = 0 \qquad \forall\ t \geq t_0, \text{ and } \quad u(t) = 0$$

Equilibrium point of LTI system

$$\dot{x} = Ax$$

- $x = 0$ is unique if A is non-singular.
- If A is singular, there will be x in null space of A. Thus, all solutions of $Ax = 0$ will be equilibrium points.

Simple Pendulum

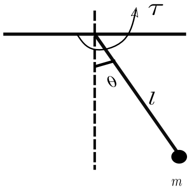


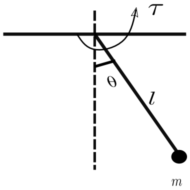
Figure: Simple Pendulum

The dynamic equation governing the equation of motion of the pendulum is given by

$$ml^2\ddot{\theta} + mgl\sin\theta = \tau$$

Consider $x_1 = \theta$ and $x_2 = \dot{\theta}$, then

Simple Pendulum



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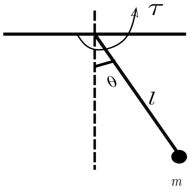
$$\begin{aligned}\dot{x}_1 &= x_2,\\ \dot{x}_2 &= -\frac{g}{l}\sin x_1 + \frac{\tau}{ml^2}\end{aligned}$$

One can write

$$\dot{x} = f(x) + g(x)u$$

Simple Pendulum



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One can write

$$\dot{x} = f(x) + g(x)u$$

where $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, $f(x) = \begin{bmatrix} x_2 \\ \frac{-g}{l}\sin x_1 \end{bmatrix}$, $g(x) = \begin{bmatrix} 0 \\ \frac{1}{ml^2} \end{bmatrix}$ and $u = \tau$.
This is affine in a control nonlinear system where $f(x)$ is a Drift vector field with $f : \mathbb{R}^2 \to \mathbb{R}^2$ and $g(x)$ is a control vector field with $g : \mathbb{R}^2 \to \mathbb{R}^2$.

Equilibrium point

$E = \{x \in \mathbb{R}^n : f(x) = 0\}$

For simple pendulum, $f(x) = 0 \implies \begin{bmatrix} x_2 \\ \frac{-g}{l} \sin x_1 \end{bmatrix} = 0$

This implies $x_2 = 0$, and $\frac{-g}{l} \sin x_1 = 0$

$x_1 = 0, \pi, 2\pi, \ldots, n\pi$

So there are two equilibrium points, $(0, 0)$, $(\pi, 0)$.

THANK YOU